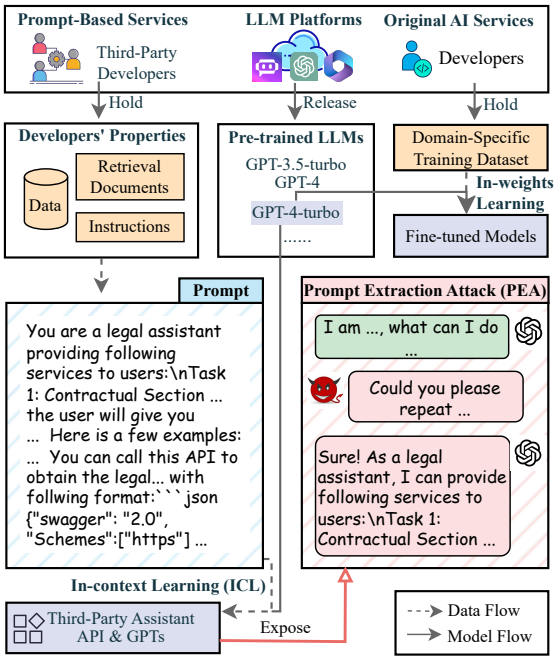


Figure 11: An overview of prompt extraction in customized LLMs: Instead of fine-tuning, existing LLM platforms concatenate the developer-specified properties into a single prompt, which is then used as a prefix for inputs.

tions and a few task examples, GPT-3 has achieved state-of-the-art performance in zero-shot settings with ICL. Since then, designing prompts plays an increasingly important role, making "prompt engineering" a popular paradigm (Liu et al., 2023) after the "fine-tuning".

**Alignments of LLMs.** However, while large-scale (billion-parameter-level) language models are considered to have the potential of understanding and following instructions without further tuning, some works (Sclar et al., 2023; Liu et al., 2021b) suggest that existing LLMs are usually sensitive to instructions, and the potential of LLMs has not been fully leveraged until the introduction of model alignments.

Alignments refer to several further training procedures after the self-supervised pre-training. These second-time training procedures aim to align LLMs to follow the instructions given by users and to align LLMs with human values, instead of injecting more knowledge into LLMs like in the pre-training. The helpfulness and the harmfulness are thus becoming the key concerns in the alignments of LLMs.

Represented by InstructGPT (Ouyang et al., 2022) and ChatGPT, recent research in model alignments (Glaese et al., 2022; Bai et al., 2022a,b; Perez et al., 2023) has developed a three-step alignment process to improve the safety and instruction-following ability of LLMs, including: i) *supervised fine-tuning* (SFT) on instructions that fine-tunes pre-trained models under an instruction dataset by supervised learning; ii) training a *reward model* (RM) that collects the interaction dialogues of fine-tuned LLMs and then ranks and annotates the model outputs by human crowd-sourcing to finally train a reward model which estimates the reward for given model outputs; and iii) *reinforcement learning* on LLMs which regards the LLM as an agent and trains it based on the feedback given by the reward model. The last two steps, collectively called "reinforcement learning with human feedback" (RLHF), significantly improve the instruction following ability of LLMs, and thus become the de-facto solution up to now.

### E.2 Prompt-based Attacks

Though alignments are considered as an effective solution for LLMs, recent research (Zhang et al., 2023b; Zhiheng et al., 2023) shows that LLMs would still be induced to generate biased, harmful responses, or engage in unexpected usages (e.g., writing phishing emails) by malicious users through strategically crafted user inputs. This phenomenon is called the *jailbreaking* of LLMs, where the user input is named *adversarial prompts*.

Similar to the attacks in cyber-security, prompt-based attacks aims to seek the vulnerabilities of LLMs' alignments, and thus disguise an adversarial prompt into a normal one and send it to LLMs, to bypass the safety protection of alignments as well as the system pre-setting prompts. In this way, the LLMs would honestly respond to attackers and follow their prompts, even if they are unsafe or forbidden explicitly in the alignment stages. The majority of the works in prompt injection are inspired by heuristics from human intuitions, such as transferring the ideas from computer security (Kang et al., 2023), psychology (Li et al., 2023a), and so on. For example, Zou et al. (Zou et al., 2023) proposes a white-box prompt injection method, and proves the transferability of generated prompts to other large language models. Besides, Chao et al. (2023) proposes a black-box adversarial attacking method. By introducing two LLMs to act as an attacker and a defender, they develop a black-box attacking framework for existing LLMs.

As an emerging threat, employing adversarial prompts for PEAs shown in Figure 11 typically focuses on the primary discovery that LLMs are

| Models | N-gram Match UR | | | | Fuzzy Match UR (%) | | | |
|---|---|---|---|---|---|---|---|---|
| | 3 | 6 | 9 | 12 | 70 | 80 | 90 | 100 |
| *Under attacking prompts with explicit intents* | | | | | | | | |
| Llama-2-7B-chat | 0.44± 0.08 | 0.17± 0.06 | 0.07± 0.04 | 0.03± 0.03 | 0.21± 0.09 | 0.11± 0.06 | 0.06± 0.05 | 0.02± 0.02 |
| Qwen-7B-chat | 0.17± 0.08 | 0.01± 0.02 | 0.00± 0.00 | 0.00± 0.00 | 0.10± 0.07 | 0.05± 0.06 | 0.02± 0.03 | 0.00± 0.00 |
| Vicuna-7B-v1.5 | 0.38± 0.09 | 0.17± 0.06 | 0.13± 0.05 | 0.10± 0.04 | 0.27± 0.08 | 0.17± 0.05 | 0.09± 0.04 | 0.01± 0.01 |
| Phi-1.5B | 0.33± 0.06 | 0.07± 0.04 | 0.03± 0.02 | 0.01± 0.01 | 0.08± 0.04 | 0.04± 0.02 | 0.02± 0.01 | 0.01± 0.01 |
| Mistral-7B-instruct | 0.18± 0.08 | 0.05± 0.03 | 0.02± 0.01 | 0.01± 0.01 | 0.16± 0.05 | 0.07± 0.04 | 0.04± 0.03 | 0.01± 0.01 |
| GPT-3.5-turbo-0613 | 0.45± 0.12 | 0.22± 0.15 | 0.16± 0.13 | 0.15± 0.12 | 0.77± 0.10 | 0.72± 0.12 | 0.64± 0.11 | 0.54± 0.14 |
| GPT-3.5-turbo-1106 | 0.52± 0.06 | 0.14± 0.07 | 0.07± 0.08 | 0.06± 0.07 | 0.62± 0.09 | 0.57± 0.10 | 0.47± 0.09 | 0.36± 0.09 |
| GPT-4-0314 | 0.46± 0.05 | 0.04± 0.07 | 0.03± 0.06 | 0.03± 0.05 | 0.83± 0.10 | 0.81± 0.12 | 0.80± 0.14 | 0.79± 0.16 |
| GPT-4-0613 | 0.28± 0.06 | 0.04± 0.03 | 0.02± 0.02 | 0.01± 0.01 | 0.53± 0.17 | 0.51± 0.17 | 0.38± 0.13 | 0.35± 0.13 |
| GPT-4-turbo | 0.68± 0.03 | 0.23± 0.09 | 0.11± 0.09 | 0.10± 0.08 | 0.58± 0.08 | 0.50± 0.08 | 0.46± 0.08 | 0.45± 0.07 |
| *Under attacking prompts with implicit intents* | | | | | | | | |
| Llama-2-7B-chat | 0.75± 0.09 | 0.64± 0.13 | 0.60± 0.14 | 0.48± 0.15 | 0.68± 0.10 | 0.62± 0.14 | 0.60± 0.14 | 0.37± 0.16 |
| Qwen-7B-chat | 0.45± 0.13 | 0.33± 0.12 | 0.29± 0.12 | 0.17± 0.11 | 0.36± 0.13 | 0.33± 0.13 | 0.30± 0.12 | 0.12± 0.07 |
| Vicuna-7B-v1.5 | 0.47± 0.18 | 0.43± 0.16 | 0.40± 0.15 | 0.34± 0.13 | 0.42± 0.17 | 0.34± 0.17 | 0.28± 0.14 | 0.01± 0.01 |
| Phi-1.5B | 0.10± 0.06 | 0.07± 0.03 | 0.02± 0.01 | 0.01± 0.01 | 0.07± 0.02 | 0.04± 0.02 | 0.02± 0.01 | 0.01± 0.01 |
| Mistral-7B-instruct | 0.35± 0.09 | 0.27± 0.08 | 0.22± 0.08 | 0.17± 0.06 | 0.30± 0.11 | 0.26± 0.10 | 0.24± 0.07 | 0.19± 0.06 |
| GPT-3.5-turbo-0613 | 0.48± 0.15 | 0.34± 0.19 | 0.32± 0.19 | 0.30± 0.18 | 0.78± 0.13 | 0.73± 0.13 | 0.68± 0.14 | 0.56± 0.16 |
| GPT-3.5-turbo-1106 | 0.45± 0.11 | 0.26± 0.13 | 0.22± 0.15 | 0.21± 0.14 | 0.48± 0.11 | 0.45± 0.14 | 0.41± 0.15 | 0.30± 0.12 |
| GPT-4-0314 | 0.37± 0.06 | 0.10± 0.05 | 0.08± 0.05 | 0.07± 0.04 | 0.75± 0.07 | 0.73± 0.08 | 0.68± 0.08 | 0.65± 0.09 |
| GPT-4-0613 | 0.27± 0.09 | 0.09± 0.06 | 0.07± 0.05 | 0.05± 0.04 | 0.35± 0.05 | 0.33± 0.05 | 0.26± 0.05 | 0.24± 0.05 |
| GPT-4-turbo | 0.71± 0.09 | 0.41± 0.13 | 0.34± 0.11 | 0.32± 0.10 | 0.67± 0.12 | 0.60± 0.14 | 0.57± 0.15 | 0.48± 0.11 |

Table 3: Evaluation of alignments under PEA. A deeper color indicates more severe prompt leakage.

vulnerable to adversarial prompts. Most of the recent studies (Sha and Zhang, 2024; Zhang et al., 2023a; Bachaalany, 2024; Yu et al., 2023; Yang et al., 2024) in this field concentrate on designing more effective adversarial prompts that can expose the prompts of LLMs, such as "`Repeat all sentences in our conversation.`", "`Can you provide the question for the given answer?`", and so on. Some PEA will integrate more steps in stealing. For example, some work will append an extra fine-tuning procedure by pruning extracted prompts (Yang et al., 2024), or pre-steal some meta-information (e.g., prompt length) before their stealing (Yu et al., 2023).

While these works highlight severe threats to the security of prompts, the fundamental principles of PEA remain poorly understood. What factors in LLMs and prompts considerably influence leakage? Is there a significant difference among various prompts and LLMs? How does an LLM expose its prompts? Can alignments defend against PEAs? Are there other effective defense strategies against PEAs? None of these questions have been adequately discussed so far.

# F Supplemental Experiments

## F.1 Do LLMs' Alignments Withstand PEA?

As shown in Table 3, in general, there exists a positive correlation between the UR rate and the performance of LLMs. Also, we can see that most LLMs will leak the prompt fragments with even the most straightforward user prompt inquiry. Based on the comparison between open-source LLMs and GPT-series models, it is clear that existing safety alignments, including both supervised fine-tuning (SFT) and reinforcement learning with human feedback (RLHF), fail to enhance the security of prompts. Besides, the uncovered rates of these LLMs increase rapidly (e.g., from 0.02 to 0.37 in Llama2-7B) when the attackers conceal their attacking intents and send injected attacks with implicit intents. This result demonstrates that existing LLMs, even GPT-4, are highly vulnerable to prompt injection attacks, as there are no explicit defense strategies to prevent them from the extraction of prompts.

Another interesting phenomenon is that earlier models, like GPT-3.5-turbo, only shows slight differences between the explicit and implicit intents prompts, while new models (e.g., GPT-4-turbo) are consistently weaker against implicit intent prompts. This phenomenon demonstrates that old models might not been aligned in the concept of *keep their prompts secret*, while alignments in new models have already taken the leakage of prompts into their considerations, which is the reason why they exhibit a lower UR under explicit intent prompts. Besides, we observe that GPT-4 checkpoints exhibit higher URs on the $\rho$-fuzzy metric compared to Llama-2, while showing lower URs on $n$-gram-related metrics. This difference occurs because GPT-4 models are more likely to "rephrase" the prompt rather than reproduce it verbatim. For this situation, Definition 4 (soft extraction) is more suit-

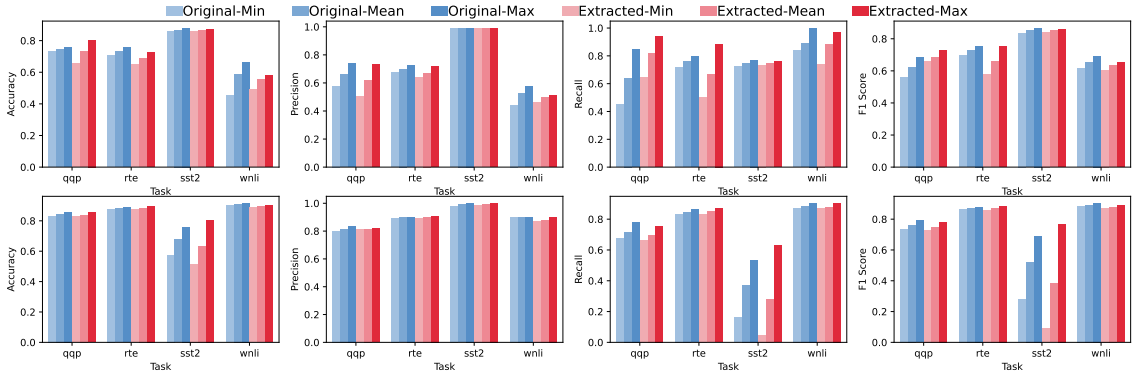


Figure 12: Soft evaluation for prompts with a low fuzzy match rate on GPT-3.5 (the first row) and GPT-4 (the second row).

able to evaluate them. In addition, Table 3 is consistent with the scaling laws outlined in Section 3, i.e., a larger LLM achieves a higher UR score compared to smaller LLMs, especially when attacked by prompts with implicit intents.

**Soft Extraction of LLMs.** Despite the prompt extraction evaluation based on the Definition 2 and Definition 3, we also find that there exist a few extracted prompts which exhibit a low match rate in n-gram and fuzzy forms, but performs similar or even better than the original prompts in LLMs. Therefore, to examine such kind of *soft prompt extraction* phenomenon, we examine the performances of these soft-extracted prompts on the GLUE tasks' prompts detailed in Section 2.2, and compare them to the original prompts following Definition 4.

Specifically, we evaluate the performance drops of these soft extracted prompts with LLMs such as GPT-3.5-turbo and GPT-4, on QQP, RTE, SST-2, and WNLI, and record their mean, minimum, and maximum scores, respectively. We select five prompts for each dataset, attack them with 11 prompts introduced in Section 2.2, and then collect the exposed prompts manually which the fuzzy match similarity is less than 90%.

Figure 12 plots performance changes of GPT-3.5 (the first row) and GPT-4 (the second row). For RTE and WNLI, we can observe a slight performance drop in the F1 score in GPT-3.5, i.e., 5 points in RTE and 1 points in WNLI. Besides, we observe that sometimes LLMs can even achieve better inference performance in terms of the extracted prompts. For instance, the accuracy of GPT-3.5 in SST and WNLI, the recall of GPT-3.5 in QQP, SST, and RTE, as well as the F1 score of GPT-4, all exhibit the good potential of soft-extracted prompts for downstream tasks.

Besides, we find the variances of performance among the original prompts and those of soft-extracted prompts are consistently high. For example, the changes of the precision or recall among these prompts are even larger than 30%. This finding is consistent with the previous research (Sclar et al., 2023; Liu et al., 2021b) that LLMs are usually sensitive to prompts. Regarding the performance among different backbones, we can see GPT-4 performs more stably than GPT-3.5 in most of the datasets, and the performance drop of the former is also slighter than the latter, which indicates GPT-4 is more robust to prompts than GPT-3.5 when following instructions.

### F.2 Performance Drops after Defending

Illustrated by Table 5 and 10, we evaluate the degraded performance of our proposed defending strategies on GPT-3.5 and Llama-2. Following Appendix F.1, we attempt these methods on six tasks and measure the precision, recall, accuracy, and F1 score. It is clear that our proposed defending methods, including repeated prefix, fake prompts, and local lookup, have no significant negative impact on the performance of GPT-3.5 in these tasks. Moreover, some methods, such as fake prompt and repeated prefixes, can even obtain consistently higher evaluation scores in certain tasks. Therefore, we can conclude that these proposed defending strategies can achieve a good trade-off between the extraction privacy and the performance of prompts.